Seeing the Air: Social Media Imagery for Seasonal Air Quality Monitoring in a Himalayan Valley City

Bhavya Saini[1], Priyanka Singh[1]
[1]Natural and Applied Sciences, TERI School of Advanced Studies

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1 Introduction and objectives

Air quality research predominantly has focused on major metropolitan regions, leaving secondary cities in topographically constrained environments underrepresented in climate and environmental informatics studies. Valley and basin cities like Dehradun exhibit pollutant trapping, particularly during post-monsoon and early winter conditions, caused by terrain barriers, atmospheric stability, calm winds and cold-air pooling (Cheng et al., 2025; Bacer et al., 2024). Further, Dehradun exemplifies a broader class of valley-constrained urban systems worldwide where topography and seasonal meteorology strongly govern pollutant dispersal. This limits horizontal and vertical air mass exchange. In addition, seasonal population intensification in the region is driven by tourism from Delhi NCT, India combined with crop-residue burning and festival activities contributes to episodic air quality deterioration. Traditional monitoring networks remain sparse in such contexts. Image-based air quality estimation using deep learning techniques to capture spatial and temporal variation in AQI categories can serve as a proxy for air quality monitoring (Kow et al., 2022; Zhang et al., 2021; Chen & Zhao, 2020; Alauddin et al., 2025) and provide comparative inter-annual insights for September–December of 2024 and 2025 in Dehradun.

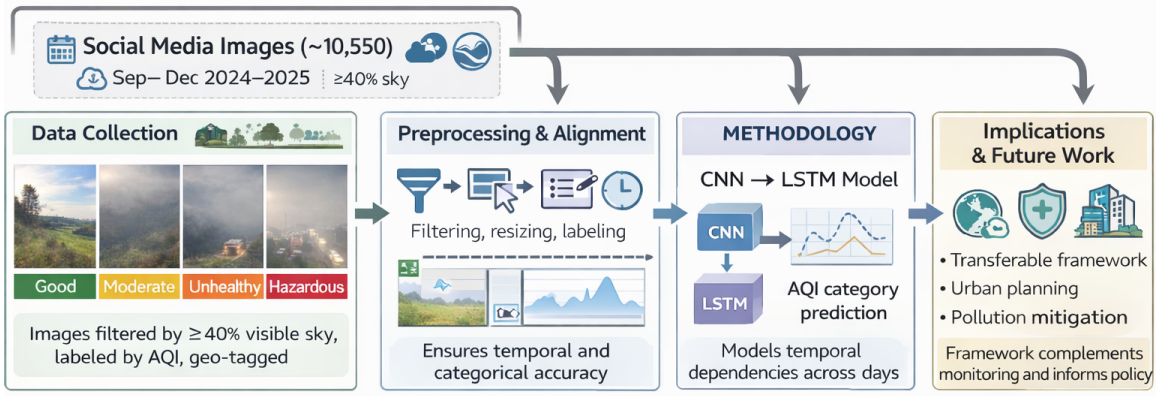

Figure 1. Methodological Workflow for Image-Based Air Quality Prediction

2 Methods and data

We curated a dataset of approximately 10,550 publicly available geo-tagged social media images from Dehradun, captured between September and December of 2024 and 2025,

filtered to retain only landscapes with at least 40% visible sky. The dataset is almost equally distributed across four AQI categories: Good, Moderate, Unhealthy and Hazardous. Each image was temporally aligned with ground-based AQI measurements from nearby monitoring stations. A CNN–LSTM framework was employed to integrate spatial and temporal information. A convolutional neural network pre-trained on ImageNet extracted visual features indicative of haze, cloud patterns and sky clarity. The long short-term memory network captured temporal dependencies across consecutive days enabling detection of episodic pollution events and seasonal trends. The analysis emphasized inter-annual comparison between 2024 and 2025 to examine the influence of tourism from Delhi NCT, crop-residue burning and festival-related emissions on air quality.

3 Preliminary results and/or expected outcomes

Preliminary inspection shows that visual sky clarity correlates with AQI categories. This indicates that the social media imagery can serve as a complementary air quality signal in valley cities. The CNN–LSTM model achieves an accuracy of 81.6% and a macro F1-score of 0.71 despite high visual variability and atmospheric ambiguity in real-world social media imagery. Classification performance is higher for extreme categories, with F1-scores of 0.85 (Good) and 0.83 (Hazardous), while moderate confusion exists between adjacent categories such as Moderate and Unhealthy. Model predictions show a strong correlation with ground-based AQI measurements ($R^2 \approx 0.71$), indicating that visual features such as haze intensity and sky clarity are reliable indicators of air quality in valley environments. Temporal analysis reveals a 22.48% increase in polluted days (Unhealthy + Hazardous) during November–December compared to earlier months, capturing seasonal pollution patterns. Inter-annual comparison shows an approximate 9.34% rise in high-pollution days in 2025 relative to 2024, reflecting variability in meteorological and anthropogenic influences. The approach offers a transferable deep learning framework for valley cities that supports climate-informed urban planning and pollution mitigation complementing traditional monitoring networks. Limitations include spatial and temporal bias, selective image filtering and separating anthropogenic from natural atmospheric variability.

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